Name = Suraj Singh

Btech CSE “B”

CU240251579

Roll no. = 59

Subject = Full Stack lab

Lab Sheet 4

#TASK 1

SUM.js

```
function sum(a, b) {
   return a + b;
} module.exports = sum;
```

Even-odd.js

```
function evenOdd(number) {
 return number % 2 === 0 ? "Even" : "Odd"; }
  module.exports = evenOdd;
```

App1.js

```
const sum = require("./sum");
const evenOdd = require("./even-odd");
console.log("Sum:", sum(10, 20));
console.log("15 is", evenOdd(15));
console.log("24 is", evenOdd(24));
```

```
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS> node
app1.js
Sum: 15
10 is: Even
7 is: Odd
```

#Task 2

file-ops.js

```
const fs = require("fs");
const fileName = "student.txt";
const newFileName = "profile.txt";
fs.writeFile(fileName, "Name: Dev Sharma\nRoll Number: CU24250269\n", (err) => {
   if (err) throw err;
```

```
 fs.appendFile(fileName, "Course: Full Stack Web Development\n", (err) => {  if (err) throw err;
 fs.readFile(fileName, "utf8", (err, data) => {   if (err) throw err;
console.log(data.trim());  fs.rename(fileName, newFileName, (err) => {
 if (err) throw err;  console.log(` Renamed ${fileName} to ${newFileName}` );
        });
      });
   });
});
```

```
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS> node
file-ops.js

File Content:
Name: Suraj Singh
Roll Number: 59
Course: Full Stack Web Development
```

#Task 3

dir-ops.js

```
const fs = require("fs");
const folder = "uploads";
const files = ["file1.txt", "file2.txt", "file3.txt"];
function createFiles() {  let remaining = files.length;   files.forEach((file) => {
 fs.writeFile(` ${folder}/${file}` , "", (err) => { if (err) throw err;
   remaining--;
if (remaining === 0) {
fs.readdir(folder, (err, entries) => { if (err) throw err;
console.log("Files:", entries.join(", ")); fs.unlink(` ${folder}/${files[0]}` , (err) => {
if (err) throw err;  console.log(` Deleted: ${files[0]}` );
}); }); }
      });
   });
}
fs.mkdir(folder, { recursive: true }, (err) => {
```

```
if (err) throw err;

createFiles(); });
```

```
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS> node
dir-ops.js
uploads folder created.
Files in uploads folder:
[ 'file1.txt', 'file2.txt', 'file3.txt' ]
file3.txt deleted.
```

#Task 4

path-demo.js

```
const path = require("path");

const filePath = "/home/user/data/report.pdf";

const relativePath = "./data/report.pdf";

console.log("Directory:", path.dirname(filePath));

console.log("Base name:", path.basename(filePath));

console.log("Extension:", path.extname(filePath));

console.log("Absolute path:", path.resolve(relativePath));
```

```
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS> node
path-demo.js
Directory name: /home/user/data
Base name: report.pdf
Extension: .pdf
Absolute path: C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-N
odeJS\data\report.pdf
```

#Task 5

sys-report.js

```
const os = require("os");

console.log("Platform:", os.platform());

console.log("Architecture:", os.arch());

console.log("Total Memory:", `${(os.totalmem() / 1024 ** 3).toFixed(2)} GB`);

console.log("Free Memory:", `${(os.freemem() / 1024 ** 3).toFixed(2)} GB`);

console.log("CPU Cores:", os.cpus().length);

console.log("Uptime:", `${(os.uptime() / 60).toFixed(2)} minutes`);
```

```
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS> node
sys-report.js
Platform: win32
Architecture: x64
Total Memory: 15.33 GB
Free Memory: 5.71 GB
CPU Cores: 12
System Uptime: 11623.13 minutes
```

#Task 6

order-events.js

```
const EventEmitter = require("events").EventEmitter;

const orderEvents = new EventEmitter();

orderEvents.on("orderPlaced", ({ id, item, amount }) => {
  console.log(` Order ID: ${id} | Item: ${item} | Amount: ₹${amount}` ); });

orderEvents.emit("orderPlaced", { id: 101, item: "Wireless Headphones", amount: 2499 });

orderEvents.emit("orderPlaced", { id: 102, item: "Smart Watch", amount: 3499 });

orderEvents.emit("orderPlaced", { id: 103, item: "Mechanical Keyboard", amount: 2999 });
```

```
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS> node order-events.js
------------------------
       ORDER RECEIPT
------------------------
Order ID: 101
Item: Laptop
Amount: ₹55000
------------------------
------------------------
       ORDER RECEIPT
------------------------
Order ID: 102
Item: Keyboard
Amount: ₹1500
------------------------
------------------------
       ORDER RECEIPT
------------------------
Order ID: 103
Item: Mouse
Amount: ₹800
------------------------
```

#Task 7

node server.js

```
const http = require("http");

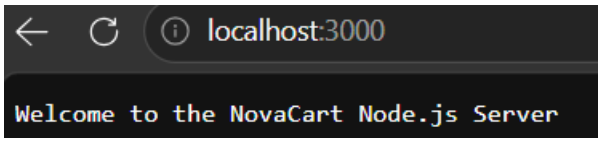
const url = require("url");

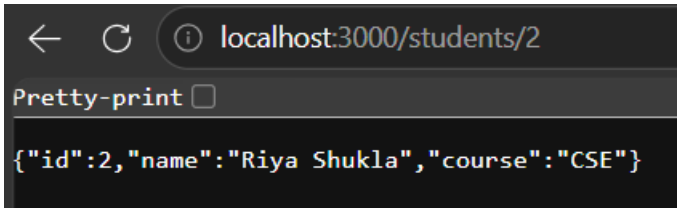
const students = [
  { id: 1, name: "Dev Sharma", course: "CSE" },
  { id: 2, name: "Riya Shukla", course: "CSE" },
```

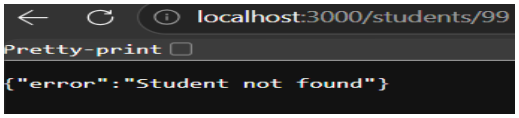
```
  { id: 3, name: "Aabra ka dabra", course: "CSE" }
];
const server = http.createServer((req, res) => {
  const parsedUrl = url.parse(req.url, true);
  const pathname = parsedUrl.pathname;
 res.setHeader("Content-Type", "application/json");
if (req.method !== "GET") {
    res.statusCode = 405;
    res.end(JSON.stringify({ error: "Method Not Allowed" }));
    return;
  }
 if (pathname === "/") {
    res.setHeader("Content-Type", "text/plain");
    res.end("Welcome to the NovaCart Node.js Server");
    return; }
 if (pathname === "/students") {  res.end(JSON.stringify(students));
    return;
  }
const studentMatch = pathname.match(/^\/students\/([^/]+)$/);
  if (studentMatch) {
    const id = Number(studentMatch[1]);
    const student = students.find((item) => item.id === id);
if (!student) {
      res.statusCode = 404;
      res.end(JSON.stringify({ error: "Student not found" }));
      return:  }
res.end(JSON.stringify(student));
    return:  }
if (pathname === "/search") {
    const keyword = parsedUrl.query.keyword || "";
    res.setHeader("Content-Type", "text/plain");
```

```
      res.end(`Search keyword: ${keyword}`);
      return; }
res.statusCode = 404;
  res.setHeader("Content-Type", "text/html");
  res.end("<h1>404 - Page Not Found</h1>");
});]
const PORT = 3000;
server.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});
```

```
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS> node
server.js
Server running at http://localhost:3000
```

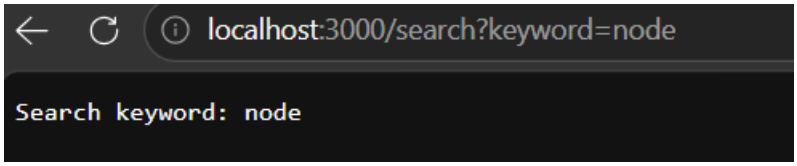






#Task 8

node server.js



#Task 9

stream-demo.js

```
const fs = require("fs");
const fileName = "large-file.txt";
```

```
let content = "";
for (let i = 1; i <= 50; i++) {
   content += `This is line ${i} of the file.\n`; }
fs.writeFile(fileName, content, (err) => {
   if (err) throw err;
 const stream = fs.createReadStream(fileName);
 stream.on("data", (chunk) => {
      console.log(`Chunk received: ${chunk.length} bytes`);
   });
stream.on("end", () => {
      console.log("Finished reading file.");
   }); });
```

```
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS> node
stream-demo.js
large-file.txt created with 50 lines.
Chunk received: 1141 bytes
Finished reading file.
```

#Task 10

id-gen.js

```
const { nanoid } = require("nanoid");
for (let i = 1; i <= 5; i++) {
   console.log(`ID ${i}: ${nanoid()}`);
}
```

```
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS> node
id-gen.js
ID 1: aNewcnXPkx6K06m1yPP1Z
ID 2: 6PnJ1-3uv4DXoEviLPU-d
ID 3: szapr4LbaTDEL0j2Z2sZS
ID 4: ceBDcJlJX3hylXzgaA2Kq
ID 5: JF23SCpYotyvvdCpRY6db
```

#Task 11

calc-cli.js

```
const [, , first, second, operator] = process.argv;
const a = Number(first);
```

```
const b = Number(second);
let result;
if (Number.isNaN(a) || Number.isNaN(b)) {
  console.log("Please enter two valid numbers.");
  process.exit(1);
}
switch (operator) {
  case "+":
    result = a + b;
    break;
  case "-":
    result = a - b;
    break;
  case "*":
    result = a * b;
    break;
  case "/":
    if (b === 0) {
      console.log("Cannot divide by zero.");
      process.exit(1);  }
    result = a / b;
    break;
  default:
    console.log("Use +, -, *, or / as the operator.");
    process.exit(1);
}
console.log(` Result: ${result}` );
```

```
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS> node calc-cli.js 10 5 +
10 + 5 = 15
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS>  node calc-cli.js 10 5 -
10 - 5 = 5
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS>  node calc-cli.js 10 5 "*"
10 * 5 = 50
PS C:\Users\HP-PC\Intership Project\Style Hub\LabSheet04-NodeJS>  node calc-cli.js 10 5 /
10 / 5 = 2
```

#Task 12

env-demo.js

```
const http = require("http");

const PORT = Number(process.env.PORT) || 4000;

const server = http.createServer((req, res) => {

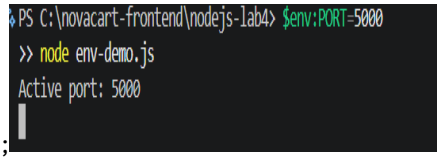
   res.writeHead(200, { "Content-Type": "text/plain" });

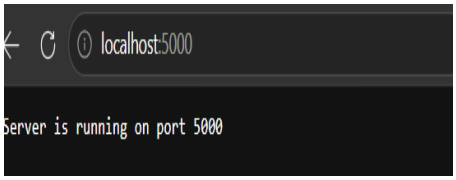
   res.end(` Server is running on port ${PORT}` );

});

server.listen(PORT, () => {

   console.log(` Active port: ${PORT}` );
```





});